$$\left.\begin{aligned} v_a &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \\ v_d &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \end{aligned}\right\} \qquad (11)$$

After second impact, a、b rest; springs are natural. At their equilibria, a、b rest and c、d move right together at v_0, uniformly, without further oscillations or collisions.

2. For $k_1 = 4k_2$

Since $k_1 = 4k_2$,

$$\omega_1 = 2\omega_2 = \sqrt{\frac{2k_1}{m}} = \sqrt{\frac{8k_2}{m}} \qquad (12)$$

For a、b、c、d, after first impact, (4) and (5) give the table-frame positions and velocities.

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{\omega_1}\sin\frac{\omega_1 t}{2} + \frac{l}{2} \end{aligned}\right\} \qquad (13)$$

$$X_d = \frac{1}{2}v_0 t - \frac{v_0}{\omega_1}\sin\frac{\omega_1 t}{2} + \frac{3l}{2}$$

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\frac{\omega_1 t}{2} \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\frac{\omega_1 t}{2} \end{aligned}\right\} \qquad (14)$$

If b and c meet in their later motion, then

$$X_b = X_c$$

From (13),

$$-\frac{1}{2}\sin\omega_1 t = \sin\left(\frac{\omega_1 t}{2}\right)$$

Since $\omega_1 = 2\omega_2$,

$$\sin\omega_2 t(\cos\omega_2 t + 1) = 0 \qquad (15)$$

The roots are

$$\omega_2 t = n\pi \qquad n = 0, \pm 1, \pm 2, \cdots \qquad (16)$$

The least positive nonzero solution t satisfies

$$\omega_2 t = \pi \qquad (17)$$

and

$$\omega_1 t = 2\pi \qquad (18)$$

b、c center-frame velocities at meeting, by (2), (16):

$$v_b'(\text{at}) = -\frac{v_0}{2}$$